

ONTOLOGY CONTENT SUMMARY

GaTO and TGC Ontologies

Classes, Hierarchies, Object Properties, Data Properties, and Controlled Values

Ontology 1: GaTO - Gamification Tutoring Ontology

Ontology 2: TGC - Teacher in Gamified E-learning Context Ontology

Prepared in English for implementation in Protege

1. GaTO - Gamification Tutoring Ontology

GaTO is an ontological model for applying gamification in Intelligent Tutoring Systems. It connects gamification concepts and theories with ITS concepts such as the domain model, student model, pedagogical model, and gamification model. The model is composed of three connected ontologies: GaDO-core, GaDO-full, and GaTO. GaDO-core contains general gamification concepts; GaDO-full extends those concepts with specific motivation theory, player models, activity loops, target behaviors, metrics, and design practices; GaTO connects those gamification concepts to ITS models and resources.

Prefixes used in this ontology: gc = GaDO-core ontology; gf = GaDO-full ontology; gt = GaTO ontology; its = reused Intelligent Tutoring Systems ontology.

1.1 Classes

```
gc:Motivation
gc:Gamification
gc:MotivationAndNeedTheory
gc:Context
gc:Game
gc:NonGame
gc:GameDesignElement
gc:Component
gc:Mechanic
gc:Dynamic
gc:Player
gc:PlayerType
gc:PlayerModel
gf:SelfDeterminationTheory
gf:BrainHex
gf:Achiever
gf:Conqueror
gf:Daredevil
gf:Mastermind
gf:Seeker
gf:Socializer
gf:Survivor
gf:MotivationalAffordance
gf:Feedback
gf:DesignPractice
gf:TargetBehavior
gf:Metric
gf:ActivityLoop
gf:EngagementLoop
gf:ProgressiveLoop
gf:Action
gf:TargetBehaviorCategory
gf:Performance
gf:Participation
gf:Competition
gf:Effectiveness
gf:Enjoyment
gf:Exploration
gt:GamifiedITS
gt:StudentModel
gt:PedagogicalModel
gt:GamificationModel
its:DomainModel
its:BehavioralKnowledge
its:InstructionalPlan
its:Curriculum
```

1.2 Class Hierarchy

```

gc:Context > gc:Game
gc:Context > gc:NonGame
gc:GameDesignElement > gc:Component
gc:GameDesignElement > gc:Mechanic
gc:GameDesignElement > gc:Dynamic
gc:MotivationAndNeedTheory > gf:SelfDeterminationTheory
gc:PlayerModel > gf:BrainHex
gc:PlayerType > gf:Achiever
gc:PlayerType > gf:Conqueror
gc:PlayerType > gf:Daredevil
gc:PlayerType > gf:Mastermind
gc:PlayerType > gf:Seeker
gc:PlayerType > gf:Socializer
gc:PlayerType > gf:Survivor
gf:ActivityLoop > gf:EngagementLoop
gf:ActivityLoop > gf:ProgressiveLoop
gf:TargetBehaviorCategory > gf:Performance
gf:TargetBehaviorCategory > gf:Participation
gf:TargetBehaviorCategory > gf:Competition
gf:TargetBehaviorCategory > gf:Effectiveness
gf:TargetBehaviorCategory > gf:Enjoyment
gf:TargetBehaviorCategory > gf:Exploration

```

1.3 Object Properties

Object Property	Domain	Range
affordsMotivation	gc:Gamification	gc:Motivation
isAppliedTo	gc:Gamification	gc:Context
makesUseOf	gc:Gamification	gc:GameDesignElement
isSatisfiedWith	gc:Gamification	gc:MotivationAndNeedTheory
interactsIn	gc:Player	gc:Context
classifiesPlayer	gc:PlayerType	gc:Player
describesType	gc:PlayerModel	gc:PlayerType
hasPlayer	gc:GameDesignElement	gc:Player
hasGameElement	gc:Player	gc:GameDesignElement
tiesComponent	gc:Component	gc:Mechanic
composesDynamic	gc:Mechanic	gc:Dynamic
usesGameDesignElements	gf:MotivationalAffordance	gc:GameDesignElement
isComposedOf	gf:EngagementLoop	gf:MotivationalAffordance
isComposedOf	gf:EngagementLoop	gf:Feedback
includes	gf:EngagementLoop	gf:Action
includes	gf:ProgressiveLoop	gf:ActivityLoop
leadsTo	gf:EngagementLoop	gf:TargetBehavior
hasMetric	gf:TargetBehavior	gf:Metric
hasCategory	gf:TargetBehavior	gf:TargetBehaviorCategory
hasTargetBehavior	gf:DesignPractice	gf:TargetBehavior
hasGameElement	gf:DesignPractice	gc:GameDesignElement
hasStudentModel	gt:GamifiedITS	gt:StudentModel
hasPedagogicalModel	gt:GamifiedITS	gt:PedagogicalModel
hasDomainModel	gt:GamifiedITS	its:DomainModel
hasGamificationModel	gt:GamifiedITS	gt:GamificationModel
hasBehavioralKnowledge	gt:StudentModel	its:BehavioralKnowledge
hasInstructionalPlan	gt:PedagogicalModel	its:InstructionalPlan
hasCurriculum	its:DomainModel	its:Curriculum

hasResources	its:Curriculum	its:Resource
hasActivityLoops	gt:GamificationModel	gf:ActivityLoop
makesUseOf	gf:Action	its:Resource

1.4 Additional Specializations Mentioned in the GaTO Paper

gc:Dynamic specializations

gc:Constraints
gc:Emotions
gc:Narrative
gc:Progression
gc:Relationships

gc:Mechanic specializations

gc:Challenges
gc:Chances
gc:Competition
gc:Cooperation
gc:Feedback
gc:ResourceAcquisition
gc:Rewards
gc:Status
gc:Story
gc:Theme
gc:Transactions
gc:Turns
gc:WinStates

gc:Component specializations

gc:Achievements
gc:Avatars
gc:Badges
gc:BossFights
gc:Collections
gc:Combat
gc:ContentUnlocking
gc:Gifting
gc:Leaderboard
gc:Levels
gc:Points
gc:Quests
gc:SocialGraph
gc:Team
gc:TimeConstraint
gc:VirtualGoods

2. TGC - Teacher in Gamified E-learning Context Ontology

TGC is an ontology for context-aware gamification recommendations for teachers. Its scope is a teachers community where teachers can share gamification experiences in terms of game elements and gamified resources. The ontology describes a teacher who wants to gamify a resource by considering teaching preferences, playing preferences,

learning-related objectives, gamification-related objectives, teacher history, learner information, and working environment. It reuses the MEMORAE / MC2 ontology and the Teacher Context Ontology, and it also uses FOAF and SIOC concepts.

The ontology is organized around three main areas: Gamified Teaching Context, Working Environment Context, and Teacher Profiling.

Prefixes used in this ontology: TGC = Teacher in Gamified E-learning Context ontology; TCO = Teacher Context Ontology; MC2 = MEMORAE / MC2 ontology; FOAF = Friend of a Friend vocabulary; SIOC = Semantically-Interlinked Online Communities vocabulary.

2.1 Classes

TGC:PlayerType
TGC:LearningStyle
TGC:GameElementResource
TGC:GamifiedResource
TGC:ClassroomObjective
TGC:Objective
TGC:PedagogicalObjective
TGC:BehaviouralObjective
TGC:EducationalTopic
TGC:SubjectArea
TGC:TopicKnowledge
TGC:DigitalCompetency
TGC:Experience
TGC:GamificationExperience
TGC:PedagogicalExperience
TGC:SocialCommitmentCompetency
TGC:DigitalContentCompetency
TCO:Learner
TCO:Lesson
TCO:Course
TCO:Audience
TCO:EducationalInstitution
TCO:WorkingEnvironment
TCO:Person
TCO:Teacher
MC2:Resource
MC2:SimpleResource
MC2:CompositeResource
MC2:ReferenceKey
MC2:IndexKey
MC2:Space
MC2:InteractionActivity
MC2:UserAccount
FOAF:Agent
SIOC:UserGroup

2.2 Class Hierarchy

TCO:Person > TCO:Teacher
MC2:Resource > MC2:SimpleResource
MC2:Resource > MC2:CompositeResource
TGC:Objective > TGC:PedagogicalObjective
TGC:Objective > TGC:BehaviouralObjective
TGC:Experience > TGC:GamificationExperience
TGC:Experience > TGC:PedagogicalExperience
TGC:DigitalCompetency > TGC:DigitalContentCompetency

2.3 Object Properties

Object Property	Domain	Range
TGC:reuseResource	TCO:Lesson	MC2:Resource
TCO:composedOf	TCO:Course	TCO:Lesson
TGC:learnsCourse	TCO:Learner	TCO:Course
TGC:isComposedOf	TCO:Lesson	TCO:Learner
TGC:hasTargetAudience	TCO:Lesson	TCO:Audience
TGC:hasLearningStyle	TCO:Learner	TGC:LearningStyle
TGC:hasPlayerType	TCO:Learner	TGC:PlayerType
TGC:hasEducationalTopic	TCO:Course	TGC:EducationalTopic
TGC:concernsACourse	TGC:ClassroomObjective	TCO:Course
TGC:communicatesObjective	TGC:ClassroomObjective	TGC:Objective
rdf:type	TGC:PedagogicalObjective	TGC:Objective
rdf:type	TGC:BehaviouralObjective	TGC:Objective
TGC:hasSubjectArea	TCO:Course	TGC:SubjectArea
TGC:examinesEducationalTopic	TGC:SubjectArea	TGC:EducationalTopic
TGC:hasTargetedTopicKnowledge	TGC:BehaviouralObjective	TGC:TopicKnowledge
TGC:relevantToObjective	TGC:GameElementResource	TGC:Objective
TGC:containsGameElement	TGC:GamifiedResource	TGC:GameElementResource
TGC:designedWithObjective	TGC:GamifiedResource	TGC:BehaviouralObjective
rdf:type	TCO:EducationalInstitution	TCO:WorkingEnvironment
TCO:worksIn	TCO:Teacher	TCO:EducationalInstitution
TCO:teaches	TCO:Teacher	TCO:Course
TCO:hasDocumentObjective	TCO:Teacher	TGC:Objective
rdf:type	TCO:Teacher	TCO:Person
rdf:type	TCO:Person	FOAF:Agent
TGC:specializedIn	TCO:Teacher	TGC:SubjectArea
TGC:hasTitle	TCO:Teacher	AcademicTitle
TGC:hasTeachingStyle	TCO:Teacher	TeachingStyle
TGC:hasFirstLanguage	TCO:Teacher	FirstLanguage
TGC:hasSecondLanguage	TCO:Teacher	SecondLanguage
TGC:hasGender	TCO:Teacher	Gender
TGC:hasName	TCO:Teacher	Name
TGC:hasAge	TCO:Teacher	Age
TGC:hasTopicKnowledge	TCO:Teacher	TGC:TopicKnowledge
TGC:hasDigitalCompetency	TCO:Teacher	TGC:DigitalCompetency
TGC:hasExperience	TCO:Teacher	TGC:Experience
TGC:hasPlayerType	TCO:Teacher	TGC:PlayerType
rdf:type	TGC:GamificationExperience	TGC:Experience
rdf:type	TGC:PedagogicalExperience	TGC:Experience
rdf:type	TGC:SocialCommitmentCompetency	TGC:DigitalCompetency
rdf:type	TGC:DigitalContentCompetency	TGC:DigitalCompetency
rdf:type	MC2:SimpleResource	MC2:Resource
rdf:type	MC2:CompositeResource	MC2:Resource
MC2:hasReference	MC2:Resource	MC2:ReferenceKey
MC2:index	MC2:ReferenceKey	MC2:IndexKey
MC2:isVisibleFor	MC2:IndexKey	MC2:Space
MC2:isReferenceKey	MC2:InteractionActivity	MC2:ReferenceKey
SIOC:hasUserGroup	MC2:Space	SIOC:UserGroup
SIOC:userGroupOf	SIOC:UserGroup	MC2:Space
SIOC:memberOf	MC2:UserAccount	SIOC:UserGroup
SIOC:hasMember	SIOC:UserGroup	MC2:UserAccount
rdf:type	MC2:UserAccount	FOAF:Agent

2.4 Data Properties and Controlled Ranges

Data Property	Range / Controlled Values	Description
TGC:hasName	xsd:string	Full name of the teacher
TGC:hasAge	xsd:integer; values from 20 to 99	Age of the teacher
TGC:hasGender	Male; Female	Gender of the teacher
TGC:hasFirstLanguage	English; French; Arabic	First preferred language for communication and teaching
TGC:hasSecondLanguage	English; French; Arabic	Second preferred language for communication and teaching
TGC:hasAcademicTitle	Instructor; Associate Professor; Professor	Academic title held by the teacher
TGC:hasSubjectArea	Social Sciences; Mathematics; Languages; Arts Education; Computer Sciences; Engineering	Domain area of the course content
TGC:hasTeachingStyle	Expert; Formal Authority; Personal Model; Facilitator; Delegator	Preferred teaching style of the teacher
TGC:hasPlayerType	Achiever; Free-spirit; Socializer; Philanthropist	Preferred player type of the teacher
TGC:hasDigitalContentCompetency	Creator; Curator	Ability or preference to create, select, organize, or adapt educational content
TGC:hasSocialCommitmentCompetency	socially limited; socially competent	Awareness and openness to use digital tools for communication and collaboration with students
TGC:hasPedagogicalExperience	<=5 years; 6-11 years; 12-17 years; 18-23 years; 24+ years	Teaching experience in years
TGC:hasGamificationExperience	Novice; Beginner; Intermediate; Advanced	Experience in understanding and applying game elements within pedagogical resources

2.5 Controlled Individuals

Class / Category	Individuals
TGC:PlayerType	Achiever; FreeSpirit; Socializer; Philanthropist
TGC:LearningStyle	ReadingWritingType; AuditoryType
TGC:DigitalContentCompetency	Creator; Curator
TGC:SocialCommitmentCompetency	SociallyLimited; SociallyCompetent
TGC:GamificationExperience	Novice; Beginner; Intermediate; Advanced
TGC:PedagogicalExperience	LessOrEqual5Years; SixToElevenYears; TwelveToSeventeenYears; EighteenToTwentyThreeYears; TwentyFourPlusYears
TeachingStyle	Expert; FormalAuthority; PersonalModel; Facilitator; Delegator
TGC:SubjectArea	SocialSciences; Mathematics; Languages; ArtsEducation; ComputerSciences; Engineering

3. Compact Implementation Lists

3.1 GaTO Compact Class List

gc:Motivation
gc:Gamification
gc:MotivationAndNeedTheory
gc:Context
gc:Game
gc:NonGame
gc:GameDesignElement

gc:Component
gc:Mechanic
gc:Dynamic
gc:Player
gc:PlayerType
gc:PlayerModel
gf:SelfDeterminationTheory
gf:BrainHex
gf:Achiever
gf:Conqueror
gf:Daredevil
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gf:Feedback
gf:DesignPractice
gf:TargetBehavior
gf:Metric
gf:ActivityLoop
gf:EngagementLoop
gf:ProgressiveLoop
gf:Action
gf:TargetBehaviorCategory
gf:Performance
gf:Participation
gf:Competition
gf:Effectiveness
gf:Enjoyment
gf:Exploration
gt:GamifiedITS
gt:StudentModel
gt:PedagogicalModel
gt:GamificationModel
its:DomainModel
its:BehavioralKnowledge
its:InstructionalPlan
its:Curriculum
its:Resource

3.2 GaTO Compact Object Property List

affordsMotivation
classifiesPlayer
composesDynamic
describesType
hasActivityLoops
hasBehavioralKnowledge
hasCategory
hasCurriculum
hasDomainModel
hasGameElement
hasGamificationModel
hasInstructionalPlan
hasMetric
hasPedagogicalModel
hasPlayer
hasResources
hasStudentModel

hasTargetBehavior
includes
interactsIn
isAppliedTo
isComposedOf
isSatisfiedWith
leadsTo
makesUseOf
tiesComponent
usesGameDesignElements

3.3 TGC Compact Class List

TGC:PlayerType
TGC:LearningStyle
TGC:GameElementResource
TGC:GamifiedResource
TGC:ClassroomObjective
TGC:Objective
TGC:PedagogicalObjective
TGC:BehaviouralObjective
TGC:EducationalTopic
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TGC:DigitalCompetency
TGC:Experience
TGC:GamificationExperience
TGC:PedagogicalExperience
TGC:SocialCommitmentCompetency
TGC:DigitalContentCompetency
TCO:Learner
TCO:Lesson
TCO:Course
TCO:Audience
TCO:EducationalInstitution
TCO:WorkingEnvironment
TCO:Person
TCO:Teacher
MC2:Resource
MC2:SimpleResource
MC2:CompositeResource
MC2:ReferenceKey
MC2:IndexKey
MC2:Space
MC2:InteractionActivity
MC2:UserAccount
FOAF:Agent
SIOC:UserGroup

3.4 TGC Compact Object Property List

MC2:hasReference
MC2:index
MC2:isReferenceKey
MC2:isVisibleFor
SIOC:hasMember
SIOC:hasUserGroup
SIOC:memberOf
SIOC:userGroupOf

TCO:composedOf
TCO:hasDocumentObjective
TCO:teaches
TCO:worksIn
TGC:communicatesObjective
TGC:concernsACourse
TGC:containsGameElement
TGC:designedWithObjective
TGC:examinesEducationalTopic
TGC:hasAge
TGC:hasDigitalCompetency
TGC:hasEducationalTopic
TGC:hasExperience
TGC:hasFirstLanguage
TGC:hasGender
TGC:hasLearningStyle
TGC:hasName
TGC:hasPlayerType
TGC:hasSecondLanguage
TGC:hasSubjectArea
TGC:hasTargetAudience
TGC:hasTargetedTopicKnowledge
TGC:hasTeachingStyle
TGC:hasTitle
TGC:hasTopicKnowledge
TGC:isComposedOf
TGC:learnsCourse
TGC:relevantToObjective
TGC:reuseResource
TGC:specializedIn
rdf:type